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Paper Code : PCC-CS503/PCCCS50 3/PCCCS503/PCCICB502 Object Oriented Programming

UPID : 005504

Time Allotted : 3 Hours

Full Marks :70

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following :

[1 x 10 = 10]

- (I) What is an Abstract Data Type (ADT)?
- (II) What is the purpose of the 'this' keyword in object-oriented programming?
- (III) What is the purpose of design patterns in software engineering?
- (IV) What is the difference between abstract classes and interfaces in object-oriented programming?
- (V) What is the purpose of the Swing library in Java?
- (VI) What is the significance of the Text example in understanding ADTs?
- (VII) How does encapsulation help in achieving data abstraction in object-oriented programming?
- (VIII) Explain the concept of encapsulation in object-oriented programming.
- (IX) Give an example of how commands can be implemented as objects in an object-oriented language.
- (X) What is the purpose of the Waterfall model in software development?
- (XI) Explain how the abstraction function can be used to ensure the correctness of an ADT implementation.
- (XII) Explain the concept of dynamic method dispatch in the context of polymorphism in object-oriented programming.

Group-B (Short Answer Type Question)

Answer *any three* of the following :

[5 x 3 = 15]

2. Write a Java program to create a simple GUI application with a button and a label. When the button is clicked, the label should display the current date and time. [5]
3. Explain the concept of information hiding in the context of ADTs, and discuss its importance in software engineering. [5]
4. Explain the different types of polymorphism in object-oriented programming with examples. Discuss the advantages and disadvantages of each type. [5]
5. Describe the concept of method overriding and method overloading in object-oriented programming. Provide examples to illustrate their differences. [5]
6. Explain the concept of data abstraction and its relationship with ADTs. Provide examples to illustrate the benefits of data abstraction in software design. <https://www.makaut.com> [5]

Group-C (Long Answer Type Question)

Answer *any three* of the following :

[15 x 3 = 45]

7. (a) Design and implement a software system that demonstrates the use of inheritance and several design patterns, such as the Iterator, Observer, and Decorator patterns. [8+7]
(b) Provide a detailed explanation of your design choices, class diagrams, and the rationale behind your implementation decisions.
8. (a) Implement a drawing application using object-oriented principles and design patterns. The application should support different shapes (e.g., rectangles, circles, lines), allow users to draw and manipulate these shapes, and provide undo and redo functionality. [9+6]
(b) Explain the object-oriented design principles and patterns used in your implementation.
9. (a) Explain the Unified Process (UP) in software development. [5+5+5]
(b) Discuss its phases, disciplines, and best practices.
(c) Provide examples of how UP can be applied to a software project.
10. (a) Analyze the time and space complexity of the operations in the ADTs you have implemented. [8+7]
(b) Identify potential areas for optimization and propose improvements to enhance efficiency.
11. (a) Discuss the importance of software documentation in the software development process. [3+6+6]

- (b) Explain different types of documentation (e.g., requirements, design, user manuals) and their purposes.
- (c) Provide examples of how documentation can be effectively developed and maintained throughout the project lifecycle.

*** END OF PAPER ***